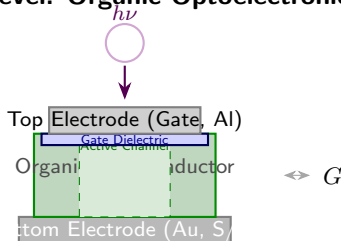
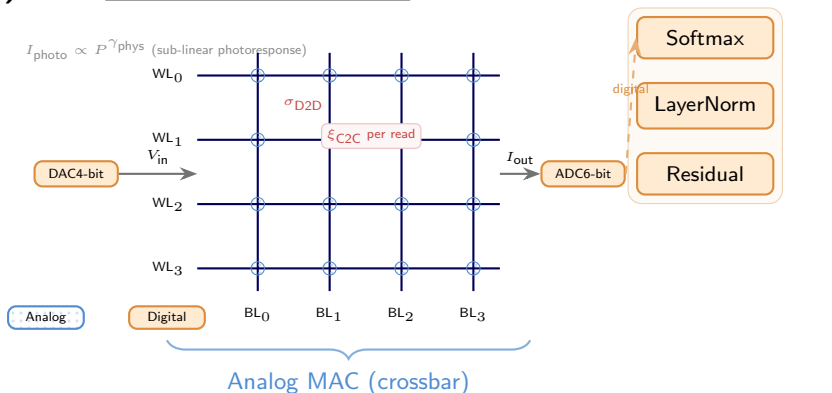


(a) Device Level: Organic Optoelectronic Memory Cell



(b) Array Level: Hybrid Analog-Digital Crossbar



(c) Network Level: ViT Hybrid Deployment Policy

TinyViT Layer	Mapping	Noise Injection
PatchEmbed (Conv2d)	Analog	D2D + C2C
QKV Projection	Analog	D2D + C2C
Attention Softmax	Digital	—
Output Projection	Analog	D2D + C2C
MLP fc1	Analog	D2D + C2C
MLP fc2	Analog	D2D + C2C
LayerNorm	Digital	—
Residual Add	Digital	—
Classifier Head	Digital	—

Dense linear ops → crossbar (analog). Dynamic/statistical ops (softmax, normalization, residual) remain digital.